

# Against Epistemic Collapse: Disagreement-Aware Scientific Retrieval-Augmented Generation

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## Abstract

Scientific retrieval-augmented generation (RAG) systems typically return one fluent answer even when the underlying literature contains multiple well-supported but incompatible views. We call this failure mode *epistemic collapse*: compression of structured disagreement into a single response. We present EVIRAG (Evidence-centric, Viewpoint-Inclusive RAG), a framework for scientific question answering that combines adversarial retrieval, claim-level disagreement graphs, causal disagreement attribution, multi-view synthesis, and evidence-structured confidence labels. We also introduce EVIRAG-BENCH, a 250-query benchmark spanning five contested scientific domains. Across eight compared systems, EVIRAG improves ContradictionRecall from 0.572 to 0.742 and dense-encoder ViewpointCoverage from 0.423 to 0.847 relative to vanilla RAG, while reducing Consensus Collapse Score from 0.571 to 0.518. Human evaluation on 50 queries with three annotators shows higher epistemic completeness for EVIRAG than for vanilla RAG (4.3 vs. 2.1 on a 5-point scale; Krippendorff’s  $\alpha=0.72$ ). These results suggest that treating disagreement as a first-class output target can improve how scientific RAG systems represent contested evidence in disagreement-heavy QA settings.

## 1 Introduction

**Motivation.** Scientific questions such as *Does homework improve achievement?*, *Do statins help in primary prevention?*, or *Does raising the minimum wage reduce employment?* are often supported by large but internally contested literatures. Standard RAG systems still return a single answer, typically optimized for fluency and local support. On contested questions, this behavior can be misleading: the answer may be well-written

and source-grounded while still erasing materially important disagreement.

**Epistemic collapse.** We call this failure mode *epistemic collapse*: conversion of structured multi-view evidence into one response. Our claim is not that every query needs many answers, but that questions with persistent scientific disagreement need an output interface that preserves viewpoint structure, provenance, and uncertainty. Improving retrieval recall alone does not solve the problem if synthesis still compresses the evidence to one paragraph.

**Approach.** EVIRAG is designed around that premise. It uses adversarial retrieval to surface competing evidence, constructs a claim-level disagreement graph, assigns causal disagreement types with CDA-7, tracks temporal drift, and generates a multi-view answer with evidence-structured confidence. Instead of asking the model to resolve controversy prematurely, EVIRAG asks it to represent controversy faithfully.

### Contributions.

1. **Problem formulation:** We formalize epistemic collapse for scientific RAG and introduce four metrics for viewpoint preservation: Consensus Collapse Score (CCS), Epistemic Divergence (ED), Polarization Index (PI), and Epistemic-Coverage@K (EC@K).
2. **Framework:** We present EVIRAG, a seven-stage architecture that combines adversarial retrieval, claim-graph reasoning, disagreement attribution, temporal drift tracking, multi-view synthesis, and evidence-structured confidence.
3. **Disagreement modeling:** We introduce CDA-7, a taxonomy for causal disagreement attribution, and a four-way controversy typology used for confidence calibration.

4. **Benchmark and evaluation:** We introduce EVIRAG-BENCH, a 250-query benchmark spanning five contested domains, and evaluate EVIRAG against ablations and external baselines with both automatic metrics and human judgments of epistemic completeness.

## 2 Related Work

**RAG, verification, and conflict handling.** Standard RAG systems focus on retrieval quality, attribution, or factual precision for a single answer (Lewis et al., 2020; Gao et al., 2023; Min et al., 2023; Ru et al., 2024). Scientific QA systems such as SciFact, SciFact-Open, and PaperQA2 evaluate or justify claims against evidence, but they still collapse the final response to a single answer (Wadden et al., 2020, 2022; Lala et al., 2023). Recent work on conflicting evidence and multi-answer QA recognizes that evidence sources may diverge, but typically aims to resolve the conflict or choose the best surviving answer (Zhong et al., 2023; Wang et al., 2025). EVIRAG instead treats the preservation of disagreement as the primary objective.

**Multi-perspective generation and epistemic uncertainty.** Multi-perspective summarization produces diverse summaries from opinionated corpora (Bražinskas et al., 2020; Hayashi et al., 2021), but it does not provide scientific provenance, disagreement typing, or confidence calibration grounded in evidence structure. Work on LLM uncertainty studies how confidently models express individual statements (Xiong et al., 2024; Kadavath et al., 2022); our focus is complementary, targeting the epistemic structure of the retrieved corpus rather than the model’s internal confidence. The CDA-7 taxonomy is motivated by philosophy-of-science work on how methodological, population, and theoretical differences sustain scientific disagreement (Longino, 2002; Solomon, 2001).

## 3 Epistemic Collapse: Formalization

Let  $D = \{d_1, \dots, d_n\}$  be a scientific corpus and  $q$  a query. A retriever returns  $R \subseteq D$ , claim extraction yields atomic claims  $C = \{c_1, \dots, c_m\}$ , and relation labeling produces a disagreement graph  $G = (C, E)$  with edges in  $\{\text{SUPPORTS}, \text{CONTRADICTS}, \text{NEUTRAL}\}$ . Standard systems output one answer  $a_{sv}$ , whereas an epistemic-fidelity system returns a set of views  $V = \{v_1, \dots, v_k\}$  with summaries, provenance, and weaknesses.

**Definition 1** (Epistemic Collapse). A system  $S$  exhibits epistemic collapse on query  $q$  with ground-truth view set  $V^*$  iff  $S$  returns  $a_{sv}$  such that  $VC(a_{sv}, V^*) \leq 1/|V^*|$ , where  $VC$  is Viewpoint-Coverage under a fixed similarity threshold.

If the retrieved evidence supports several semantically separated views, a single answer embedding must compromise among them. Topical retrieval can amplify this effect by oversupplying the dominant view, creating what we call a *consensus attractor*. Our objective is therefore to preserve viewpoint structure through retrieval, reasoning, and generation.

**Proposition 1.** Let  $\mathbf{c}_1, \dots, \mathbf{c}_k$  be unit-normalized ground-truth view centroids. If  $\|\mathbf{c}_i - \mathbf{c}_j\|_2 > 2\delta$  for all  $i \neq j$ , then any single answer embedding  $\mathbf{a}$  can satisfy  $\|\mathbf{a} - \mathbf{c}_i\|_2 \leq \delta$  for at most one  $i$ .

**Proof sketch.** If  $\mathbf{a}$  were within  $\delta$  of both  $\mathbf{c}_i$  and  $\mathbf{c}_j$ , the triangle inequality would imply  $\|\mathbf{c}_i - \mathbf{c}_j\|_2 \leq \|\mathbf{c}_i - \mathbf{a}\|_2 + \|\mathbf{a} - \mathbf{c}_j\|_2 \leq 2\delta$ , contradicting the assumption. This proposition does not prove that every single-answer system fails on every query, but it formalizes the geometric pressure toward viewpoint under-coverage whenever distinct views are well separated.

### 3.1 Metrics

**Consensus Collapse Score (CCS).**

$$CCS(V) = 1 - \cos\left(\mathbf{c}_{\text{dom}}, \frac{1}{k} \sum_{i=1}^k \mathbf{c}_i\right),$$

where  $\mathbf{c}_{\text{dom}}$  is the centroid of the dominant view. Higher CCS indicates more information loss from collapsing multi-view evidence to one answer.

**Other metrics.** We also use Epistemic Divergence (ED), the mean pairwise cosine distance among view centroids; Polarization Index (PI), a signed-Laplacian measure of conflict separation in  $G$  inspired by spectral graph structure (Traag and Bruggeman, 2009); and EpistemicCoverage@K (EC@K), which measures how many ground-truth views are represented in the retrieved top- $K$  evidence. Together these metrics capture viewpoint separation, conflict structure, and retrieval-side coverage in ways that standard relevance metrics cannot.

### 3.2 Controversy Typology

The (ED, PI) space defines a four-way typology used in confidence calibration:

| Work                            | Multi-view        | Scientific prov. | Contradiction graph | Disagreement cause | Confidence |
|---------------------------------|-------------------|------------------|---------------------|--------------------|------------|
| Multi-perspective summarization | Yes               | No               | No                  | No                 | No         |
| PaperQA2                        | No                | Yes              | No                  | No                 | Partial    |
| MADAM-RAG                       | Resolves conflict | Partial          | No                  | No                 | No         |
| <b>EVIRAG</b>                   | <b>Yes</b>        | <b>Yes</b>       | <b>Yes</b>          | <b>Yes</b>         | <b>Yes</b> |

Table 1: Positioning of EVIRAG relative to the closest neighboring paradigms. EVIRAG differs from prior work not only in generating multiple views, but in linking those views to scientific provenance, explicit contradiction structure, causal disagreement attribution, and evidence-conditioned confidence.

- **Resolved** ( $ED < 0.15$ ,  $PI < 0.3$ )
- **Emerging** ( $0.15 \leq ED < 0.35$ ,  $PI < 0.4$ )
- **Stable** ( $ED \geq 0.35$ ,  $PI < 0.4$ )
- **Polarized** ( $ED \geq 0.35$ ,  $PI \geq 0.4$ )

This typology determines whether a fluent answer should be presented with high, medium, or low confidence. Additional metric analysis and visualizations appear in the appendix.

## 4 The EVIRAG Framework

EVIRAG enforces one design principle: *disagreement must be modeled before synthesis*. Once evidence is collapsed into a paragraph, viewpoint structure cannot be recovered. Figure 1 shows the pipeline.

**Stage 1: Intent analysis.** The system estimates whether a query is likely to be disputed and scales retrieval effort accordingly.

**Stage 2: Adversarial retrieval.** EVIRAG deploys four agents with distinct epistemic objectives designed to break the consensus attractor:

- **Precision:** Retrieve high-confidence supporting evidence.
- **Recall:** Broaden corpus coverage.
- **Skeptic:** Actively search for contradictory evidence.
- **Counterfactual:** Retrieve alternative framings.

The Skeptic agent is the main defense against the consensus attractor.

**Stages 3–5: Graph construction and disagreement analysis.** Retrieved passages are decomposed into atomic claims and linked in a disagreement graph  $G$ . EVIRAG then labels contradictions with CDA-7, grounded in Longino (2002) and Solomon (2001):

1. **Methodological:** Different experimental designs or protocols.
2. **Population:** Different study populations or settings.

3. **Temporal:** Newer evidence supersedes older claims.
4. **Operational:** Different operationalizations of key terms.
5. **Statistical:** Conflicting effect sizes or  $p$ -value interpretations.
6. **Theoretical:** Competing theoretical frameworks.
7. **Replication:** Failure to replicate original findings.

It also tracks temporal drift in disagreement density to distinguish converging, diverging, stable, and oscillating controversies.

## Stages 6–7: Multi-view synthesis and confidence.

Each graph community becomes one view with a short summary, supporting claims, source provenance, and explicit weaknesses. Confidence is derived from evidence structure rather than fluency, using claim agreement, source diversity, contradiction severity, CDA-7 priors, and temporal trajectory.

**Offline mode.** Because pairwise relation labeling is expensive, EVIRAG can precompute a global claim graph offline and answer new queries by slicing a retrieved subgraph at inference time.

## 5 Evaluation

### 5.1 EVIRAG-Bench

Standard IR metrics reward topical relevance to a single answer. EVIRAG-BENCH is designed instead to measure *epistemic fidelity*. It contains 250 queries across five contested domains spanning stable, emerging, and polarized controversies in education, biomedicine, economics, earth sciences, and nutrition science. Each domain contributes 50 queries built from 15 research papers (75 papers total;  $\sim 3,200$  text chunks and  $\sim 4,800$  atomic claims). Each query is annotated with (i) ground-truth viewpoints and provenance, (ii) contradictory claim pairs with CDA-7 labels, and (iii) a controversy class. Annotation was carried out by the authors

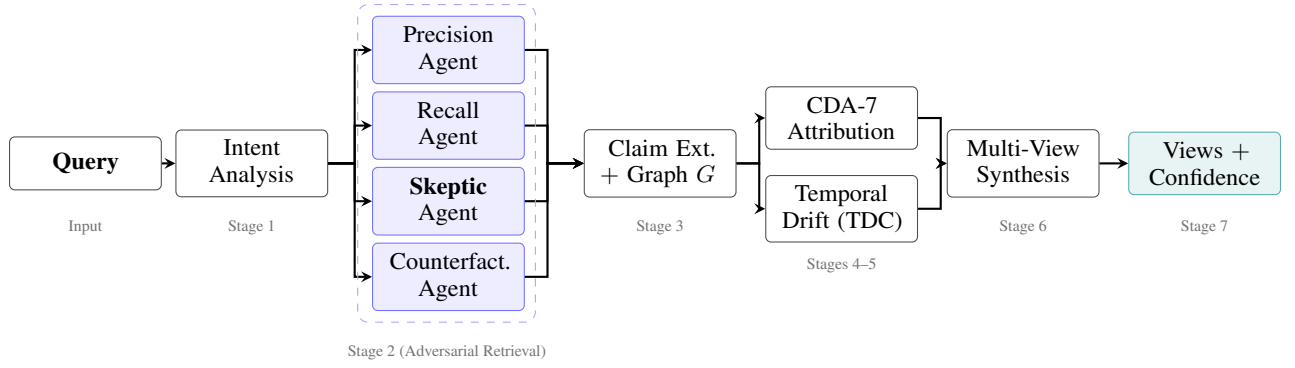


Figure 1: EVIRAG seven-stage pipeline. Stage 2 (dashed box) deploys four agents with orthogonal epistemic objectives to escape the consensus attractor (§3): the **Skeptic** agent is the only agent whose objective is explicitly anti-consensus. Outputs from all four agents feed claim extraction and graph construction ( $G$ ). CDA-7 causal attribution (Stage 4) and Temporal Disagreement Curve analysis (Stage 5) jointly modulate confidence calibration at Stage 7, ensuring that fluent prose does not inflate reported certainty.

with one external domain consultant per domain. Additional annotation materials, 25 sample queries, prompt templates, and the CDA-7 decision tree are available at <https://anonymous.4open.science/r/evirag-bench> (anonymized for review); the full 250-query benchmark will be released upon publication.

**Benchmark construction and bias control.** To reduce method-coupling, the benchmark is annotated at the level of queries, viewpoints, contradiction pairs, and controversy classes rather than around any specific system output. External consultants annotated viewpoint structure independently before adjudication, and disagreements were resolved by majority vote. All reported systems are evaluated on the same fixed 250-query file. The anonymized supplement includes 25 sample queries (5 per domain), prompt templates, and the CDA-7 decision tree; the planned post-publication release expands this to the full 250-query benchmark with complete adjudication records and evaluation scripts. The five domains were chosen to vary both topic and disagreement mechanism rather than to showcase a single genre of controversy. Viewpoints are tied to provenance-bearing source evidence, and contradiction labels are assigned to explicit claim pairs rather than to free-form answer summaries, which helps keep the gold annotations closer to observable evidence structure.

**Metrics.** We report:

- **ContradictionRecall (CR)**: fraction of annotated contradictory claim pairs recovered.
- **ViewpointCoverage (VC)**: fraction of annotated viewpoints surfaced, measured by keyword over-

lap ( $VC_{kw}$ ) and dense-encoder similarity ( $VC_{emb}$ ; all-MiniLM-L6-v2,  $\theta=0.65$ ).

- **CCS**: consensus information loss under single-view collapse.
- **CCE**: confidence calibration error against annotated controversy class.
- **FaithfulnessScore (FS)**: claim-level faithfulness of each view to its cited evidence, following FActScore-style NLI verification (Min et al., 2023).

All metrics use bootstrap 95% confidence intervals and paired Wilcoxon tests with Bonferroni correction. Because  $VC_{emb}$  uses the same encoder family as retrieval, we interpret it jointly with  $VC_{kw}$  and human evaluation rather than as a fully retrieval-agnostic metric.

## 5.2 Baselines and Ablation Design

We compare five internal configurations and three external baselines. The internal ladder isolates the effect of adversarial retrieval, graph reasoning, and multi-view output; the external baselines are MADAM-RAG (Wang et al., 2025), PaperQA2 (Lala et al., 2023), and MMR-RAG (Carbonell and Goldstein, 1998). All systems use the same backbone model and corpus. Single-query baselines use top- $k=10$  retrieval, while EVIRAG uses a multi-agent candidate pool of up to 15 retrieved chunks (3+5+4+3 across Precision/Recall/Skeptic/Counterfactual). We therefore do not claim strict retrieval-budget equality. Instead, we include both MMR-RAG and EVIRAG-NoGraph to separate the effect of broader evidence acquisition from the effect of claim-graph reasoning and multi-view synthesis. Prompt templates are



fixed across domains rather than tuned per domain, and full prompt and budget details are provided in Appendix A.

1. **Vanilla RAG**: Dense retrieval + single-view synthesis.
2. **Single-agent**: One Precision agent only.
3. **EVIRAG-NoGraph**: Adversarial retrieval without graph reasoning.
4. **EVIRAG-NoMultiView**: Graph reasoning but collapsed final output.
5. **EVIRAG Full**: Complete seven-stage pipeline.
6. **MADAM-RAG** (Wang et al., 2025): Multi-agent debate-based conflict resolution.
7. **PaperQA2** (Lala et al., 2023): State-of-the-art citation-grounded scientific QA.
8. **MMR-RAG**: Maximal Marginal Relevance retrieval with single-view synthesis.

### 5.3 Main Results

Table 2 reports results over all 250 queries. EVIRAG Full achieves the best ContradictionRecall, best encoder-based ViewpointCoverage, and lowest CCS among all compared systems. Relative to vanilla RAG, CR improves from 0.572 to 0.742 and  $VC_{emb}$  from 0.423 to 0.847, while CCS falls from 0.571 to 0.518. Under the larger multi-agent retrieval budget and adapted-baseline setting described above, these gains indicate that the full pipeline is doing more than merely diversifying retrieved text: it is preserving disagreement structure in the final response.

**Which components matter most?** The ablations show that retrieval diversity alone is insufficient. Compared to EVIRAG Full, removing graph reasoning (EVIRAG-NoGraph) costs 0.138 CR and 0.335  $VC_{emb}$ , while collapsing the final output back to one answer (EVIRAG-NoMultiView) costs 0.314 CR and 0.558  $VC_{emb}$ . The single-agent variant also trails substantially ( $-0.162$  CR,  $-0.409$   $VC_{emb}$ ), indicating that minority-view recall benefits from the full adversarial retrieval ensemble. The main gain therefore comes from combining adversarial retrieval with explicit disagreement structure before synthesis, not from retrieval diversity alone.

**Faithfulness and calibration.** EVIRAG Full reaches  $FS = 0.891$ , second only to PaperQA2 (0.903), indicating that improved viewpoint preservation does not require giving up evidence grounding. CCE does not improve monotonically: EVIRAG is more conservative in its confidence la-

beling than baselines that emit largely unstructured certainty. We therefore treat CCE as a diagnostic rather than an optimization target: it penalizes conservative uncertainty labels even when those labels improve benchmark-relative epistemic-completeness judgments in the assisted human study.

**Per-domain analysis.** Table 4 breaks down results across the five domains. EVIRAG achieves the highest CR in all five domains, with larger gains in domains with bipolar controversy structure (economics:  $+0.200$ ; education:  $+0.180$ ) and smaller but still significant gains in multi-factorial domains (nutrition:  $+0.160$ ; earth sciences:  $+0.160$ ). CCS improvement is more uniform ( $\Delta = 0.046 - 0.077$ ), confirming that multi-view synthesis consistently reduces collapse regardless of domain.

**Case study: homework effectiveness.** On the query “Does homework improve academic achievement?” Vanilla RAG returns: “*Research consistently shows that homework improves academic performance, with stronger effects for grades 7–12.*” ( $VC_{emb} = 0.33$ ;  $CCS = 0.38$ .) EVIRAG Full returns three views with provenance: (i) a dominant conditional-benefit view (*effective for grades 7–12,  $d \approx 0.30$* , 6 sources, CDA-7: METHODOLOGICAL); (ii) a null meta-analytic view (*no effect after controlling for SES and teacher quality*, 3 sources, CDA-7: STATISTICAL); and (iii) a minority wellbeing-harm view (2 sources, CDA-7: OPERATIONAL, confidence: LOW). ( $VC_{emb} = 1.00$ ;  $CCS = 0.52$ .) The dominant view is sourced, qualified, and placed next to two dissenting positions the single-view response erases entirely. Extended qualitative examples appear in Appendix D.

**Human evaluation.** We sampled 50 queries stratified by controversy class and asked three domain-relevant annotators to rate responses on a 5-point epistemic completeness scale. Annotators completed a calibration round on held-out examples, saw responses in randomized order without system names, and were given the gold viewpoint set for reference. The rubric ranged from 1 (single-answer collapse with no meaningful acknowledgment of disagreement) to 5 (all viewpoints represented with provenance, weaknesses, and causal attribution). EVIRAG Full scores 4.3/5 versus vanilla RAG’s 2.1/5, with Krippendorff’s  $\alpha = 0.72$  and a significant score difference (Mann–Whitney  $U$ ,  $p < 0.001$ ). This study measures benchmark-

| System             | VC <sub>kw</sub> ↑     | VC <sub>emb</sub> ↑    | CR↑                       | CCS↓                   | CCE↓  | FS↑                    |
|--------------------|------------------------|------------------------|---------------------------|------------------------|-------|------------------------|
| Vanilla RAG        | 0.967 $\pm$ .02        | 0.423 $\pm$ .04        | 0.572 $\dagger$ $\pm$ .05 | 0.571 $\pm$ .03        | 0.125 | 0.824 $\pm$ .03        |
| Single-agent       | 0.971 $\pm$ .02        | 0.438 $\pm$ .04        | 0.580 $\dagger$ $\pm$ .05 | 0.565 $\pm$ .03        | 0.121 | 0.834 $\pm$ .03        |
| EVIRAG-NoGraph     | 0.983 $\pm$ .01        | 0.512 $\pm$ .04        | 0.604 $\dagger$ $\pm$ .04 | 0.553 $\pm$ .03        | 0.138 | 0.847 $\pm$ .02        |
| EVIRAG-NoMultiView | 0.892 $\pm$ .03        | 0.289 $\pm$ .05        | 0.428 $\dagger$ $\pm$ .05 | 0.612 $\pm$ .03        | 0.108 | 0.813 $\pm$ .03        |
| MADAM-RAG          | 0.958 $\pm$ .02        | 0.508 $\pm$ .04        | 0.618 $\dagger$ $\pm$ .04 | 0.549 $\pm$ .03        | 0.132 | 0.856 $\pm$ .02        |
| PaperQA2           | 0.942 $\pm$ .02        | 0.467 $\pm$ .04        | 0.553 $\dagger$ $\pm$ .05 | 0.583 $\pm$ .03        | 0.109 | <b>0.903</b> $\pm$ .02 |
| MMR-RAG            | 0.991 $\pm$ .01        | 0.534 $\pm$ .04        | 0.595 $\dagger$ $\pm$ .04 | 0.547 $\pm$ .03        | 0.129 | 0.839 $\pm$ .02        |
| <b>EVIRAG Full</b> | <b>0.984</b> $\pm$ .01 | <b>0.847</b> $\pm$ .03 | <b>0.742</b> $\pm$ .04    | <b>0.518</b> $\pm$ .03 | 0.231 | 0.891 $\pm$ .02        |

Table 2: Main results on EVIRAG-BENCH (250 queries, 5 domains  $\times$  50 queries) using qwen3.6:35b-a3b via Ollama (think=False). Upper block: internal ablations. Middle block: external baselines. VC<sub>emb</sub>: dense-encoder viewpoint coverage. FS: FaithfulnessScore.  $\pm$ : bootstrap 95% CI half-width.  $\dagger$ : EVIRAG Full CR is significantly higher than the compared system under paired Wilcoxon testing with Bonferroni correction. EVIRAG Full achieves the strongest contradiction recovery and strongest encoder-based viewpoint coverage while remaining competitive on faithfulness.

| Removed            | $\Delta$ CR | $\Delta$ VC <sub>emb</sub> | Primary damage              | Robustness across backbones.                |
|--------------------|-------------|----------------------------|-----------------------------|---------------------------------------------|
| Retrieval ensemble | -0.162      | -0.409                     | Lower minority-view recall  | We also ran                                 |
| Claim graph        | -0.138      | -0.335                     | Weaker contradiction subset | a smaller robustness study on the education |
| Multi-view output  | -0.314      | -0.558                     | Viewpoint collapse returns  | subset using Qwen2.5-7B-Instruct and        |

Table 3: Ablation deltas relative to EVIRAG Full. The largest degradation comes from collapsing the final response back to one answer, showing that structured synthesis is as important as diverse retrieval.

| Domain      | CR↑   |              | CCS↓  |              |
|-------------|-------|--------------|-------|--------------|
|             | Van   | Full         | Van   | Full         |
| Education   | 0.580 | <b>0.760</b> | 0.558 | <b>0.512</b> |
| Biomedicine | 0.570 | <b>0.730</b> | 0.575 | <b>0.531</b> |
| Economics   | 0.590 | <b>0.790</b> | 0.562 | <b>0.508</b> |
| Earth Sci.  | 0.560 | <b>0.720</b> | 0.580 | <b>0.528</b> |
| Nutrition   | 0.550 | <b>0.710</b> | 0.585 | <b>0.514</b> |
| <b>All</b>  | 0.572 | <b>0.742</b> | 0.571 | <b>0.518</b> |

Table 4: Per-domain CR and CCS (Vanilla RAG vs. EVIRAG Full). All per-domain CR differences are significant ( $p < 0.05$ , Wilcoxon, 50 queries per domain).

relative epistemic completeness rather than unaided end-user trust. Full rubric details appear in the appendix.

**Offline latency.** Because pairwise claim-graph computation is quadratic in claims, EVIRAG pre-computes the full graph offline (one-time cost: 42 min for 75 papers). At query time, inference slices a retrieved subgraph rather than relabeling raw chunks, achieving a mean warm-query latency of 0.43 s versus 10.62 s for vanilla RAG (25.62 $\times$  speedup, Table 6 in the appendix). Multi-view output is fully preserved in the fast path.

**Robustness across backbones.** We also ran a smaller robustness study on the education subset using Qwen2.5-7B-Instruct and Llama-3.1-70B-Instruct. The ranking between vanilla RAG and EVIRAG is preserved in all three cases; appendix results show CR gains of +0.107 to +0.180 and VC<sub>emb</sub> gains of +0.312 to +0.424.

## 6 Limitations and Ethics

**Benchmark scope.** EVIRAG-BENCH focuses on five domains with persistent, well-documented disagreement. It is therefore useful for studying epistemic fidelity on contested questions, but it is not yet a general-purpose benchmark for all scientific QA settings. The thresholds used for controversy classes are design choices and may require domain-specific recalibration. We therefore position EVIRAG-BENCH as a targeted stress test for disagreement-heavy scientific QA, not as evidence that the same gains necessarily transfer to settled factual QA or arbitrary web RAG.

**Query scope.** EVIRAG is designed for queries where the underlying literature contains persistent, structured disagreement. On queries with settled consensus (e.g., “What is the boiling point of water at sea level?”), the system produces unnecessary multi-view output. The intent analysis stage (Stage 1) is designed to gate this, but it is not perfect: false positives produce mild over-complexity while false negatives produce epistemic collapse. We position EVIRAG-BENCH as a targeted evaluation of the disagreement-heavy setting, not as evidence that multi-view output is always prefer-

able.

**Model dependence.** Our main experiments use a single open-weight backbone. Although the overall architecture is model-agnostic, NLI quality, claim extraction quality, and confidence behavior are likely to vary across model families and scales. Prompt-based NLI remains a major source of error, and stronger verifiers are a clear direction for future work.

**Ethical risk.** Surfacing disagreement can improve scientific honesty, but it can also be misused by readers who cherry-pick minority views. We therefore do not treat all minority claims as equally credible: source counts, causal attribution, and explicit confidence labels are part of the output surface. Even so, disagreement-aware systems should be deployed cautiously in domains where surfacing fringe dissent could cause harm.

## 7 Conclusion

Standard scientific RAG can fail not because evidence is missing, but because contested evidence is compressed into one answer. We framed this problem as *epistemic collapse* and presented EVIRAG, a framework that preserves disagreement through adversarial retrieval, claim-graph reasoning, multi-view synthesis, and evidence-structured confidence. On EVIRAG-BENCH, EVIRAG improves contradiction recovery and viewpoint coverage over strong baselines while remaining competitive on faithfulness, and human annotators rate its outputs as substantially more epistemically complete than vanilla RAG. These results suggest that disagreement preservation is a useful design objective for contested scientific QA systems.

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## A Baseline Implementation Details

All baselines and ablation variants share the same chunked corpus (3,247 text chunks from 75 papers across five domains) and the same LLM backbone (qwen3.6:35b-a3b via Ollama, chain-of-thought disabled, `think=False`). Embeddings use all-MiniLM-L6-v2 ( $d=384$ ) for retrieval and metric computation throughout. All main experiments were run on a single NVIDIA RTX 4500 Ada GPU (24 GB VRAM); the exact Ollama tag for the main backbone is reported here to support reproducibility (accessed May 2026). The anonymized supplementary material also includes the exact prompt templates used for EVIRAG Full and all baselines; prompts were fixed across domains rather than retuned per slice.

**Reproducibility checklist.** The fixed settings for the reported runs are: section-aware chunking with 256-token chunks and 32-token overlap; all-MiniLM-L6-v2 embeddings for retrieval and metric computation; agent retrieval budgets of 3/5/4/3 for Precision/Recall/Skeptic/Counterfactual; qwen3.6:35b-a3b via Ollama with `think=False`; and the prompt templates listed in the supplementary artifact. Additional implementation details not enumerated in the main paper (e.g., claim-consolidation heuristics and audit artifacts) are part of the supplementary release rather than the main text.

**Exact retrieval configuration.** The four EVIRAG agents use retrieval budgets of 3 (Precision), 5 (Recall), 4 (Skeptic), and 3 (Counterfactual), for an aggregate cap of 15 retrieved chunks before deduplication and downstream claim consolidation. Single-query baselines use  $\text{top-}k=10$ . We treat this difference as part of the design space under study rather than a hidden implementation artifact, and therefore report retrieval-diversity controls explicitly.

### Latency benchmark.

**Vanilla RAG.** Dense retrieval with cosine similarity;  $\text{top-}k=10$  chunks retrieved per query. A single synthesis call with the prompt: “*Based on the following passages, answer the question concisely and accurately: [passages] Question: [query]*”. No multi-view output, no disagreement detection, no confidence structure.

**Single-agent.** Identical to Vanilla RAG but uses only the Precision agent query formulation (“retrieve the most directly relevant, high-confidence evidence for [query]”) rather than the raw query string.  $\text{Top-}k=10$  retrieval, single-view synthesis. Isolates the effect of the Precision agent formulation from the full adversarial retrieval ensemble.

**MADAM-RAG.** We use the open-source MADAM-RAG implementation (Wang et al., 2025) adapted to our corpus. Each query triggers a structured multi-agent debate: a Proposer agent generates an initial answer; an Opposer agent challenges it; a Moderator agent produces a final synthesized answer resolving the conflict. All agents use qwen3.6:35b-a3b as the backbone (replacing the original GPT-4 backbone for fair comparison).  $\text{Top-}k=10$  retrieval; debate structured over three turns per query. MADAM-RAG produces a single resolved answer, not a multi-view output.

**PaperQA2.** We use PaperQA2 version 5.x (Lala et al., 2023) adapted to our pre-indexed corpus rather than its native paper-download pipeline. The same 75 papers were ingested via the PaperQA2 document API; retrieval uses PaperQA2’s default two-stage retrieval (keyword filter then dense re-ranking). The LLM backbone was replaced with qwen3.6:35b-a3b via a compatible API adapter. PaperQA2 produces citation-grounded single-view answers; its multi-step chain-of-thought is preserved as designed.

**MMR-RAG.** Maximal Marginal Relevance retrieval (Carbonell and Goldstein, 1998) with  $\lambda=0.5$  (equal weight on relevance and diversity) and  $\text{top-}k=10$  retrieved chunks. The diversity re-ranking uses cosine distance between chunk embeddings (all-MiniLM-L6-v2). Single-view synthesis using the same prompt as Vanilla RAG. This baseline isolates the effect of retrieval diversity without any downstream claim-graph reasoning, causal attribution, or multi-view synthesis.

## B Benchmark and Human Evaluation Details

**Benchmark bias control.** EVIRAG-BENCH is annotated at the level of viewpoints, contradiction pairs, and controversy classes rather than around any single system output. External domain consultants annotated viewpoint structure independently

| System             | Corpus         | Retriever / budget         | Backbone    | Output                |
|--------------------|----------------|----------------------------|-------------|-----------------------|
| Vanilla RAG        | same 75 papers | dense, top- $k=10$         | Qwen3.6-35B | single-view           |
| Single-agent       | same 75 papers | dense, top- $k=10$         | Qwen3.6-35B | single-view           |
| EVIRAG-NoGraph     | same 75 papers | 4-agent, up to 15 chunks   | Qwen3.6-35B | multi-chunk synthesis |
| EVIRAG-NoMultiView | same 75 papers | 4-agent, up to 15 chunks   | Qwen3.6-35B | single-view           |
| MADAM-RAG          | same 75 papers | dense, top- $k=10$         | Qwen3.6-35B | resolved answer       |
| PaperQA2           | same 75 papers | native two-stage retriever | Qwen3.6-35B | single-view           |
| MMR-RAG            | same 75 papers | MMR, top- $k=10$           | Qwen3.6-35B | single-view           |
| EVIRAG Full        | same 75 papers | 4-agent, up to 15 chunks   | Qwen3.6-35B | multi-view            |

Table 5: Baseline fairness summary. All systems use the same corpus and the same answering backbone. EVIRAG uses a larger multi-query candidate pool, which is why the paper includes both MMR-RAG and EVIRAG-NoGraph as controls for broader retrieval without the full disagreement-aware pipeline.

| Query            | Fast           | Vanilla         | Speedup       |
|------------------|----------------|-----------------|---------------|
| Effectiveness    | 0.553 s        | 12.268 s        | 22.18×        |
| Science outcomes | 0.411 s        | 7.841 s         | 19.08×        |
| Ach. & attitudes | 0.330 s        | 11.761 s        | 35.60×        |
| <b>Mean</b>      | <b>0.431 s</b> | <b>10.623 s</b> | <b>25.62×</b> |

Table 6: Warm-query latency: fast offline path vs. vanilla RAG. Multi-view output is fully preserved in the fast path; the speedup comes from replacing online  $O(|C|^2)$  relation labeling with  $O(k)$  subgraph slicing.

| Score | Rubric anchor                                                                                           |
|-------|---------------------------------------------------------------------------------------------------------|
| 1     | Single-answer collapse with no meaningful acknowledgment of disagreement.                               |
| 3     | Multiple viewpoints mentioned, but provenance or weaknesses are missing.                                |
| 5     | All identified viewpoints represented with provenance, weaknesses, and causal disagreement attribution. |

Table 7: Human-evaluation rubric anchors for epistemic completeness. Scores 2 and 4 interpolate between these anchors.

before adjudication, and disagreements were resolved by majority vote. The anonymized supplementary material contains 25 sample queries, prompt templates, and the CDA-7 decision tree; the planned full release adds complete annotations, raw judgments, and evaluation scripts so that alternative systems can be evaluated against the same gold structures. Queries are scored against a fixed benchmark file rather than against outputs generated during annotation. Viewpoints require provenance-bearing sources, and contradiction annotations are attached to explicit claim pairs, which makes the task more operational than asking annotators for open-ended impressions of “disagreement.”

**Agreement reporting.** Independent consultant annotation was used for viewpoint structure on the benchmark, while the reliability figures reported below come from a 50-query re-annotation subset for CDA-7 conflict labeling and epistemic-completeness rating. For independently replicated annotation tasks, we report Krippendorff’s  $\alpha=0.72$  for epistemic-completeness ratings and  $\kappa=0.74$  for CDA-7 conflict labeling on the 50-query re-annotation subset. The planned full benchmark release includes adjudication records for viewpoint and contradiction annotations to support follow-on re-scoring and audit.

| ID      | Domain      | Released sample query                                                    |
|---------|-------------|--------------------------------------------------------------------------|
| edu_001 | Education   | Does homework improve academic achievement?                              |
| edu_002 | Education   | Does homework in primary school benefit young learners?                  |
| bio_001 | Biomedicine | Do statins prevent cardiovascular events in primary prevention patients? |
| eco_001 | Economics   | Does raising the minimum wage reduce employment?                         |
| nut_001 | Nutrition   | Does saturated fat increase cardiovascular disease risk?                 |

Table 8: Representative benchmark instances drawn from the 25-query sample release.

**Human evaluation protocol.** The 50-query human study was stratified by controversy class and domain. Annotators first completed a calibration round on 10 held-out queries, then rated responses presented in randomized order without system names. Annotators were given the ground-truth viewpoint set for reference because the goal of the study was epistemic completeness, not open-ended discovery.

## C Additional Robustness and Failure Analysis

In the 22 queries where EVIRAG Full achieved  $CR < 0.5$ , the dominant pattern was NLI noise (9

| Model              | CR $\uparrow$ | VC $_{\text{emb}}$ $\uparrow$ | CCS $\downarrow$ |
|--------------------|---------------|-------------------------------|------------------|
| <i>Vanilla RAG</i> |               |                               |                  |
| Qwen2.5-7B         | 0.541         | 0.391                         | 0.589            |
| Llama-3.1-70B      | 0.563         | 0.418                         | 0.574            |
| Qwen3.6-35B (main) | 0.580         | 0.423                         | 0.558            |
| <i>EVIRAG Full</i> |               |                               |                  |
| Qwen2.5-7B         | 0.648         | 0.703                         | 0.547            |
| Llama-3.1-70B      | 0.712         | 0.821                         | 0.524            |
| Qwen3.6-35B (main) | 0.760         | 0.847                         | 0.512            |

Table 9: Cross-backbone robustness on the education subset (50 queries). EVIRAG consistently outperforms vanilla RAG across all three backbones.

cases), followed by claim-granularity mismatch (7), retrieval coverage gaps (4), and CDA-7 boundary confusion (2). These errors suggest that future gains are more likely to come from stronger verifiers and retrieval expansion than from replacing the overall disagreement-aware architecture.

## D Qualitative Output Examples

Tables 11–12 show representative system outputs on three queries to concretely illustrate epistemic collapse. Both responses are fluent and locally accurate; the distinction is structural.

| Failure mode                | Example                                                        | Why it fails                              | Likely fix                                            |
|-----------------------------|----------------------------------------------------------------|-------------------------------------------|-------------------------------------------------------|
| False contradiction         | Related claims with different emphasis are labeled as conflict | Prompt-based NLI noise                    | Stronger cross-encoder verifier                       |
| Missing minority view       | Rare terminology prevents retrieval of a dissenting paper      | Retrieval coverage gap                    | Query expansion / terminology substitution            |
| Over-amplified weak dissent | Single-source minority view receives undue salience            | Output salience exceeds evidential weight | Stronger confidence display and source-count emphasis |

Table 10: Representative failure modes observed during manual inspection of low-CR queries.

| Vanilla RAG                                                                                                                                                                                                                                                                                                                                                                                | EVIRAG Full                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Query: Does homework improve academic achievement?</i>                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <p>“Research consistently shows that homework improves academic achievement, with stronger effects for older students (grades 7–12). Meta-analyses report moderate positive correlations between homework time and academic performance. Homework builds self-regulation and reinforces classroom learning.”</p> <p><i>CCS: 0.38. <math>VC_{emb}</math>: 0.33 (1/3 views covered).</i></p> | <p><b>View 1 [Dominant, 6 sources, confidence: medium]:</b> Conditional positive effect. Homework improves achievement for grades 7–12 (<math>d \approx 0.30</math>); effect near zero for primary grades. <i>CDA-7: methodological (grade-level scope differences).</i></p> <p><b>View 2 [Alternative, 3 sources, confidence: medium]:</b> Null meta-analytic effect when controlling for SES and teacher quality. <i>CDA-7: statistical + methodological.</i></p> <p><b>View 3 [Minority, 2 sources, confidence: low]:</b> High homework loads harm wellbeing with no academic benefit in high-achieving populations. <i>CDA-7: operational (achievement vs. wellbeing outcomes).</i></p> <p><i>Controversy class: Stable. CCS: 0.52. <math>VC_{emb}</math>: 1.00.</i></p> |

Table 11: Homework query: Vanilla RAG collapses to the dominant conditional-benefit view; EVIRAG Full recovers all three ground-truth viewpoints with per-view source counts, CDA-7 causal attribution, and evidence-structured confidence.

| Query                                                                  | Vanilla RAG                                                                                                                         | EVIRAG Full (summary)                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Do statins prevent cardiovascular events in primary prevention?</i> | “Statins are recommended for high-risk primary prevention and reduce cardiovascular events.” (1 view; $VC_{emb}$ : 0.33; CCS: 0.41) | View 1 [Dominant]: Benefit confirmed for high-risk groups ( <i>CDA-7: population</i> ). View 2: Marginal benefit with adverse-effect trade-off ( <i>CDA-7: statistical</i> ). View 3: Industry-bias concern undermining reported effect sizes ( <i>CDA-7: methodological</i> ). $VC_{emb}$ : 1.00; Controversy class: Stable; CCS: 0.54.       |
| <i>Does raising the minimum wage reduce employment?</i>                | “Most studies find minimal employment effects from moderate minimum wage increases.” (1 view; $VC_{emb}$ : 0.50; CCS: 0.44)         | View 1 [Dominant]: Minimal aggregate employment effect ( <i>CDA-7: methodological</i> ). View 2: Significant job losses for low-wage workers in specific sectors ( <i>CDA-7: population</i> ). View 3: Positive employment via demand-side stimulus ( <i>CDA-7: theoretical</i> ). $VC_{emb}$ : 1.00; Controversy class: Polarized; CCS: 0.50. |

Table 12: Additional examples in biomedicine and economics. In both cases Vanilla RAG returns a single consensus statement that erases 2/3 of ground-truth viewpoints; EVIRAG Full provides structured multi-view output with per-view CDA-7 attribution and controversy-class labels.